

IDRINSKOYE, Russian Federation

WMO: 297660

Lat: 54.365N

Lon: 92.150E

Elev: 925

StdP: 14.21

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.2	-31.2	-41.2	0.5	-36.9	-35.4	0.8	-31.1	13.7	23.7	11.7	23.8	1.6	110	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	86.7	67.5	83.5	66.1	80.1	64.7	69.8	82.3	67.9	79.5	66.2	77.1	4.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	98.1	74.7	63.9	92.0	72.7	62.3	86.8	71.3	34.5	82.4	32.9	79.3	31.5	77.1	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.5	9.0	-38.0	93.0	7.3	3.8	-43.3	95.7	-47.6	97.9	-51.7	100.0	-57.0	102.7		
(5)			-37.0	73.0	7.6	1.9	-42.5	74.4	-46.9	75.5	-51.2	76.6	-56.8	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.1	-3.2	2.3	19.9	38.1	50.5	63.0	66.3	61.8	49.3	36.1	18.6	4.0
	DBStd	26.46	16.63	15.42	13.25	9.12	8.69	6.88	5.08	6.30	7.45	8.61	13.41	15.59
	HDD50	7292	1648	1335	933	365	102	3	0	2	102	435	942	1425
	HDD65	11476	2113	1755	1398	806	454	116	45	138	473	896	1392	1890
	CDD50	1478	0	0	0	10	118	392	506	367	81	4	0	0
	CDD65	186	0	0	0	0	5	55	86	38	2	0	0	0
	CDH74	2456	0	0	0	4	141	829	941	489	52	0	0	0
	CDH80	692	0	0	0	0	31	253	267	128	13	0	0	0
	WSAvg	3.5	2.6	2.7	3.4	4.5	4.8	3.9	3.4	3.4	3.6	3.8	3.6	2.9
	PrecAvg	15.1	0.5	0.5	0.4	0.7	1.3	2.2	2.4	2.6	1.8	1.1	0.9	0.8
(15) Precipitation	PrecMax	18.7	1.0	1.4	1.0	1.5	3.3	3.7	4.6	6.4	4.2	3.0	1.5	1.9
	PrecMin	10.6	0.1	0.1	0.0	0.0	0.4	0.1	0.9	1.1	0.4	0.4	0.2	0.2
	PrecStd	2.0	0.2	0.3	0.2	0.4	0.7	0.9	0.9	1.3	0.7	0.5	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	36.1	51.5	71.9	84.5	90.7	90.1	88.3	80.5	64.2	48.9	35.4
		MCWB	33.1	31.5	41.2	53.0	61.2	66.6	69.6	68.9	62.0	51.0	42.8	30.6
	2%	DB	30.5	32.4	45.0	65.0	78.4	86.4	86.5	83.7	72.4	56.8	43.5	31.6
		MCWB	27.4	28.9	37.4	50.0	59.0	66.3	68.5	67.1	59.3	47.5	37.8	28.2
	5%	DB	25.7	29.0	41.4	59.6	73.8	82.8	83.3	79.6	67.4	52.3	39.0	28.2
		MCWB	23.0	25.9	35.4	47.5	56.5	64.5	67.3	65.5	56.2	45.5	35.0	25.2
	10%	DB	21.4	25.0	38.0	54.3	69.0	79.0	79.6	75.6	62.8	48.7	35.7	25.0
		MCWB	19.0	22.4	33.1	44.5	54.5	63.3	65.9	63.6	53.9	43.3	32.7	22.9
	0.4%	WB	33.5	32.4	41.4	55.3	63.9	70.2	73.0	71.4	63.7	52.1	43.3	31.5
		MCDB	37.1	35.2	49.5	71.4	80.8	84.9	84.3	83.9	77.0	60.3	48.4	34.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.7	29.2	38.0	51.2	60.7	68.0	70.6	68.7	60.1	48.8	38.3	28.4
		MCDB	30.3	32.4	44.1	63.2	74.5	81.5	82.5	80.6	70.3	55.2	42.4	31.4
	5%	WB	23.0	26.0	35.7	48.0	58.0	66.1	68.7	66.5	57.7	46.2	35.4	25.4
		MCDB	25.5	28.9	40.8	58.3	70.9	79.6	79.8	77.2	65.5	52.0	38.5	27.9
	10%	WB	19.0	22.4	33.3	45.1	55.5	64.4	67.0	64.4	55.1	43.6	32.7	23.0
		MCDB	21.3	25.1	37.7	53.9	66.8	76.8	77.1	73.4	61.2	48.2	35.7	25.1
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	19.8	21.3	20.8	23.2	23.2	20.9	20.8	20.4	16.7	14.1	15.3
		MCDBR	14.5	17.0	21.9	31.6	33.1	31.8	28.2	28.6	29.3	24.0	13.9	13.8
		MCWBR	12.6	14.2	15.1	18.4	16.4	13.6	12.2	13.3	15.7	14.9	10.1	11.7
		MCDBR	14.4	16.5	20.1	28.7	28.5	27.1	24.0	25.1	26.2	22.6	12.9	13.6
	5% WB	MCWBR	12.7	13.9	14.1	17.5	14.9	12.1	11.3	12.2	14.3	14.5	9.6	12.0
(40) Clear-Sky Solar Irradiance	taub		0.286	0.322	0.370	0.412	0.422	0.410	0.423	0.411	0.372	0.354	0.325	0.291
	taud		2.026	1.985	1.927	2.066	2.155	2.296	2.280	2.283	2.325	2.234	2.056	1.969
	Ebn at Noon		197	231	246	256	261	266	259	255	250	221	177	161
	Edh at Noon		23	34	45	46	44	39	39	36	31	27	23	20
(44) All-Sky Solar Radiation	RadAvg		305	629	1067	1379	1608	1892	1683	1405	963	527	266	197
	RadStd		30	46	101	80	185	184	122	121	112	51	35	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page